PROBLEM STATEMENT:

Mass gatherings frequently experience unpredictable pedestrian surges that can rapidly escalate into fatal crushes. Existing surveillance methods remain dangerously reactive and fail to accurately calculate physical crowd pressure in high-density environments. This vulnerability is severely compounded during mass emergencies when cellular and internet infrastructures inevitably collapse, severing critical communication between field staff and control rooms. There is an urgent need for a localized, offline-capable crowd management solution that proactively monitors crowd flow and forecasts imminent congestion before critical thresholds are breached and dynamically reroutes pedestrians toward unblocked exits through automated, on-the-ground audio-visual alerts, ensuring continuous and coordinated emergency response entirely independent of external networks. Furthermore, this system must deliver robust emergency support by instantly localizing physical distress signals, dynamically rerouting pedestrians toward unblocked paths, and automating critical on-ground interventions—such as unlocking emergency exits and dispatching first responders—ensuring a continuous, coordinated rescue effort entirely independent of external networks.